SHASHWAT KUSHWAHA | 21IM3EP29

B.Tech.(Hons.) in INDUSTRIAL ENGINEERING and MASTER OF ENGINEERING ENTREPRENEURSHIP
MICRO SPL. in ARTIFICIAL INTELLIGENCE AND APPLICATIONS, ENTREPRENEURSHIP AND INNOVATION

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2026	M.TECH Dual Degree 5Y	IIT Kharagpur	7.89 / 10
2021	CLASS XII CBSE	ABHINAV VIDYALAYA	84%
2019	CLASS X CBSE	LUCKNOW PUBLIC SCHOOL	97.2%

INTERNSHIPS

Atomberg Technologies Pvt. Ltd | Data Analyst Intern | Pune **May'25 - Jul'25**
***Objective**: To solve a NP-hard supply chain problem with data analysis and optimisation algorithms and deliver measurable efficiency gains*
- Identified **22%** volumetric wastage via multi-level Pareto on **1,500+ SKUs**; linked inefficiency to **170+** non-standard box sizes, enabling corrective action
- Achieved a **83% predictive accuracy** by designing a **heuristic** algorithm for complex 3D pallet-packing and validated against **150+** operational test cases
- Built a **Streamlit** dashboard for over **3,500** bin locations; integrated **SQL** metrics and predictive alerts to proactively prevent critical out-of-space events
- Projected **30%** cut in the pallet volume wastage and enabled multi-vendor part commonisation, driving long-term efficiency & cross-supplier optimization

University of Kent | Deep Learning Research Intern | United Kingdom **May'24 - Jul'24**
***Objective:**Conduct a comparative study between the RAG and Representation Fine - Tuning (ReFT) to enhance information retrieval*
- Created a specialized **NLP** dataset by scraping over **50** privacy policies using **Python/bs4**; leveraging key insights from review of over **30** academic papers
- Drove **10% uplift** in retrieval performance by architecting and deploying advanced **RAG** pipeline leveraging sentence window retrieval with the **GroqAPI**
- Achieved a **10%** performance gain over **RAG** baseline by applying **Representation Fine-Tuning (ReFT)** to refine the **Llama3-7B** model's core parameters

PROJECTS

Automated AI-Driven Interview Platform | Bachelor Thesis Project | Prof. Mrigank Sharad **Jan'25 - Mar'25**
***Objective:** Create a complete, AI-driven interview system using LLMs to mitigate human bias and deliver a truly scalable recruitment solution*
- Delivered AI interview application by integrating **Groq API + Llama 3** with **LlamaIndex** for optimized RAG data pipelines, architected a **4-**layer platform
- Improved **retrieval accuracy >10%** on **MRR@10** by applying sentence-window retrieval with cross-encoder, engineered a robust two-stage RAG pipeline
- Ensured reliable AI assessments with **91% faithfulness, 88% precision &85% relevancy** verified on **50+ DeepEval** tests, rigorously substantiated platform

QuakeAide: AI Assistant for Disaster Insights | Course Project | Prof. Mamata Jenamani **Jan'24 - Apr'24**
***Objective:**To deploy stateless RAG and stateful conversational AI systems, providing access to disaster data via distinct Flask applications*
- Enabled faster disaster intelligence via **Flask + GPT-3** web apps, integrating **LangChain** for real-time stream analysis, architected &deployed at scale
- Improved retrieval accuracy &relevance with **OpenAIEmbeddings** + **ChromaDB** vector index, engineered preprocessing pipeline for disaster datasets
- Strengthened decision-making with **RetrievalQA + ChatOpenAI** from **LangChain**, applied to **500+** docs, designed dynamic, efficient access workflows
- Delivered scalable AI disaster services with **RESTful API** on cloud servers, optimized to handle **250+ weekly queries**, launched &maintained reliably

Predictive Analytics for Patient Health Scoring | Self Project **Jan'24- Feb'24**
***Objective:**To build predictive & generative AI tools to convert patient SDOH data into accurate risk scores and actionable guidance*
- Engineered a high-accuracy scoring system (**R^2 = 0.6510**) with XGBoost & LightGBM, achieving low error (**MAE 9.212**, **RMSE 4.246**) on patient SDOH data
- Maximized model performance via rigorous forward selection, training **37 XGBoost** models to identify and optimize with the **15** most impactful predictors
- Developed an innovative **GPT-4** few-shot tool to translate complex SDOH features into actionable healthcare guidance, demonstrating LLM application

COMPETITION/CONFERENCE

Adobe Behaviour Simulation Challenge | Gold | Inter IIT Tech Meet 12.0
- Managed a significant **13%** RMSE improvement on a **300,000**-tweet dataset by training a custom MLP on **150,000** rows with a stabilizing SAM optimizer
- Outperformed **13x** larger **GPT** model by engineering a routing system for **3** fine-tuned LLMs (**13B** & **7B**) with a hybrid retriever for prompt augmentation
- Ensured model scalability by devising **PCA**-based method ensembling predictions from the top **10** similar users to forecast engagement for new profiles
- Confirmed superiority by benchmarking against **6** alternative approaches, including **Graph Neural Networks**, **ViT-MAE**, and gradient boosting (**XGBoost**)

SKILLS AND EXPERTISE

Languages &Libraries Python | R | C++ | C | SQL | Pandas | Numpy | Matplotlib | Seaborn | Boto3 | OpenCV | BeautifulSoup | SpaCy | Plotly | Pytest
Frameworks LangChain | PyTorch | TensorFlow | scikit-learn | Llama_Index | PySpark | FastAPI | Ragas | pyreft | Optuna | HuggingFace | Detectron2
Developer Tools VS Code | Docker | Azure |Git | Github |Anaconda | Jupyter Labs | Kaggle | Google Colab | PyCharm | Codeblocks | SageMaker

COURSEWORK INFORMATION

Machine Learning Foundations and Applications | Recommender Systems and Applications | Statistical Learning with Applications + Lab | Probability and Statistics | Linear Algebra | Advanced Calculus | Programming and Data Structures | Operations Research | Supply Chain Management | Simulation + Lab Production Planning and Control | Quality Design Control | Optimisation Heuristics Methods + Lab | Management of Inventory Systems | System Design

POSITIONS OF RESPONSIBILITY

Hall President | Vidyasagar Hall of Residence | Indian Institute of Kharagpur **Aug'24-Aug'25**
- Drove strategic academic policy reform for **16,000+** students as Student Senator, translating hall-level feedback into actionable campus-wide initiatives
- Elected to lead **26-member council**, orchestrating strategic initiatives for **300+** residents that substantially boosted engagement in hall welfare and events
- Oversaw **INR 1.8M+** budget to deliver portfolio of **40+** annual events, measurably increasing resident engagement and fostering a supportive community

Core Team Head | Finance and Logistics | Optima 2023 | IIT Kharagpur **Aug'23 - Apr'24**
- Spearheaded sponsorship, publicity, and event strategy as **Founding Member** for the inaugural **Optima** fest, driving execution to attract **1,000+** attendees
- Directed a budget exceeding **INR 1.1 million** and secured strategic partnerships with **15+** industry leaders, including Optym, Air India, and Delhivery

!Self declared by the student, CDC could not verify the relevant documents